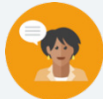


4.2 Identifying Key Features of Lines From Graphs

Lesson Introduction

 Benchmark	MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function, and determine and interpret its key features.		 Learning Target	I can identify key features of a linear function from a graph.
 Benchmark Coherence	Building On MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.		Working Toward MA.912.F.1.7 Compare key features of two functions each represented algebraically, graphically, in tables or written descriptions.	
 Essential Questions	- What are the key features of linear functions that can be determined from a graph? - What are the similarities and differences of the key features when the line is horizontal or vertical?			
 Lesson Narrative	Students begin identifying key features in this lesson, with key features limited to x-intercept, y-intercept, domain, range, and slope. Students will explore the graph of a linear function to discover these key features and identify them from different representations of lines. This lesson is Part 1 of a two-part exploration on key features, with students' instruction focused on the discovery and identification stages of key features.			
 Component Summary	4.2.1 Warm-Up (~10 min.)	“Notice and Wonder” about an equation and function in two different forms (<i>SE p. 169</i>)	4.2.3 Exploration (15 min)	Key features framed through the lens of horizontal and vertical lines (<i>SE p. 173</i>)
	4.2.2 Exploration (20 min)	Students explore a line segment to discover key features (<i>SE p. 170</i>)	4.2.4 Wrap-Up (~5 Min)	Students engage in self-reflection (<i>SE p. 174</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> x-intercept y-intercept <u>Additional Terms:</u> - Domain - Range - Slope		N/A	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may switch the domain and range while reading the graph - Students may struggle with applying the differences between positive and negative slope - Students may mix up which value is 0 when writing x- and y-intercepts	- Activities 4.2.2 and 4.2.3 are Exploration activities. Students should be allowed to go through the discovery process of this lesson to explore the key features to be able to later identify and interpret them - The 4.2.3 Exploration provides an opportunity for class discussion on key features of horizontal and vertical lines
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 4.2.1 Warm-Up provides an opportunity for students to discuss what they notice about equations and functions in two different forms. Students also discuss what they wonder about similar outputs. Try a similar process for vertical and horizontal lines in 4.2.3 Exploration.	MA.K12.MTR.2.1: Multiple Representations This lesson provides opportunities for students to make connections between key features and the way a function or equation is represented mathematically. Particularly 4.2.3 Exploration, positions students to make connections between the range as the y -intercept and the domain as the x -intercept.
 Differentiate Instruction	Consider what collaboration looks and sounds like for 4.2.2 and 4.2.3 Explorations. Opportunities to work with partners, discuss findings, and refine mathematical responses are intentionally embedded in the lesson to increase differentiated entry points for learning. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

4.2.1 Warm-Up: What Do You Notice?

Component Overview: The Warm-Up has students explore domain and range (written as input and output) while using both equation and function notation.



4.2.1 Warm-Up: What Do You Notice?

1. What do you notice about the equation $y + 7 = \frac{5}{4}(x - 8)$ and the function $f(x) = \frac{5}{4}x - 1$?

Answers will vary. Sample Answer: I notice that one equation is written in function notation and the other is not. I notice that both equations have 5/4 but the other values are different.

2. If you substitute the same x -value, or input, into the equation and function, do you think that the equation and function will have the same outputs given the same input values?

Answers will vary. I think that the equation and the function will give different outputs because they have different values. I think they will give different answers because when the same input for x is placed into each, they will give two different answers or outputs.

3. What do you notice about the equation $y - 3 = -6(x + 5)$ and the function $f(x) = -6x - 27$?

Answers will vary. I notice that one is a function and one is an equation. I notice that the equation is in point-slope form and requires distribution and the function does not.

4. Do you think the equation and function will have the same outputs given the same input values?

Answers will vary. Students do not need to write an explanation here.



Students can engage independently in a Notice and Wonder routine for questions 1-4 before they prove equivalence by substitution for questions 5-7. Although both questions portray the equations as possibly equivalent, only the second set are equivalent. Watch for the common misconception where students may write $y + 7 = \frac{5}{4}(x - 8)$ as $y = \frac{5}{4}x - 1$ because they fail to distribute the $\frac{5}{4}$ to the -8 .



Notice and Wonder

Consider this as a Notice and Wonder instructional routine, with questions 1 and 3 being “Notice” and questions 2 and 4 being “Wonder” for student responses. Students do not need to identify whether the equation and function are equivalent but ask questions that position them to prove equivalence in questions 5-7.



Why do you think that the equations are written differently?

4.2.1 Warm-Up: What Do You Notice? (continued)

5. Substitute the x -value in the equation and the function to determine if the output is the same.

x -value	Equation	Function
-2	$y + 7 = \frac{5}{4}(x - 8)$	$f(x) = \frac{5}{4}x - 1$
	$y + 7 = \frac{5}{4}(-2 - 8)$	$f(-2) = \frac{5}{4}(-2) - 1$
	$y + 7 = \frac{5}{4}(-10)$	$= \frac{10}{4} - 1$
	$y + 7 = -\frac{50}{4}$	$= -\frac{14}{4} \text{ or } -\frac{7}{2}$
	$y = -\frac{50}{4} - 7$	$\text{or } -3\frac{1}{2}$
9	$y - 3 = -6(x + 5)$	$f(x) = -6x - 27$
	$y - 3 = -6(9 + 5)$	$f(9) = -6(9) - 27$
	$y = -6(14) + 3$	$= -54 - 27$
	$= -84 + 3$	$= -81$
	$= -81$	

6. What is one observation you have about the equation and function that do **not** have the same output?
Answers will vary. I noticed that when the equation was simplified it did not match the function. I noticed when using the same input the equation and function resulted in different output values.
7. What is one observation you have about the equation and function that do have the same output?
Answers will vary. I noticed that when the equation was simplified, it matched the function. I noticed when using the same input the equation and function resulted in the same output values.



Collect and Display

If executing a Notice and Wonder routine for this activity, consider capturing students' contributions on chart paper to use later in Lesson 3. The connections students make between forms of a line might be helpful scaffolding for 4.3.3 Guided Instruction in Lesson 3.



This activity could extend to take up more time than intended for students who understand conceptually but struggle to calculate. Consider partnering students with different strengths to split the work to manage the time. The focus of the lesson is not the evaluation itself. This can, however, provide valuable intervention data.

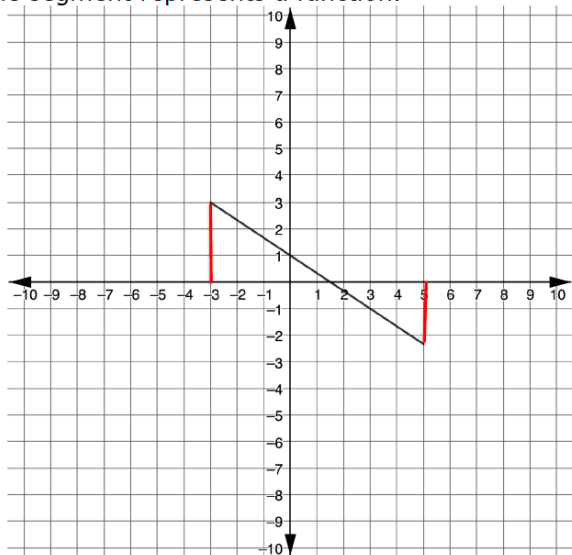
4.2.2 Exploration: Key Features From a Graph

Component Overview: Students explore key features of line segment and a line to compare the key features.



4.2.2 Exploration: Key Features From a Graph

1. A line segment is shown on the coordinate plane. The line segment represents a function.



- a. Draw a vertical line from the left endpoint of the line segment to the x -axis. What is the x -value?
-3
- b. Draw a vertical line from the right endpoint of the line segment to the x -axis. What is the x -value?
5
- c. Discuss with your partner how finding the x -values of the left endpoint and the right endpoint of the line segment helps to determine the domain of the function the line segment represents.
The endpoints give the smallest possible value and the largest possible value.



Use formal and informal observations to determine whether students can engage in this Exploration independently or if teacher support is needed. Question 1a-b position students to discover the domain and range using points on the graph. Students then formalize these points as the domain (questions 1c-d) and range (questions 1e-f) using symbols of inequality. Students should then work in pairs or small groups to identify other key features in questions 1g-h before applying this process again in practice for question 2. Students have heard words like 'domain' and 'range' before and will have varying levels of mastery. Be careful not to frontload information or give away the learning target – this exploration is intentionally scaffolded for students to connect informal findings (questions 1a-b) with formal vocabulary (questions 1c-f) and applied practice (question 2g-h). The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.



Sentence stems for discussion can include formal vocabulary, but do not need to – this discussion is intentionally scaffolded to provide entry points without that vocabulary.

4.2.2 Exploration: Key Features From a Graph (continued)

- d. Use the x -values of the left endpoint and the right endpoint of the line segment to write the domain as an inequality.

$$\underline{-3} \leq x \leq \underline{5}$$

- e. Work with your partner to determine the range of the function that is represented by the line segment.

Answers may vary between $-2\frac{1}{3}$ to $-2\frac{1}{2}$ for the lower value. $-2\frac{1}{3} \leq y \leq 3$

- f. Use the y -values of the left endpoint and the right endpoint of the line segment to write the range as an inequality.

$$\underline{-2\frac{1}{3}} \leq y \leq \underline{3}$$

- g. Work with your partner to find the other key features of the line segment.

- The slope of the line segment is $-\frac{2}{3}$.
- As the x -values are increasing, the y -values are decreasing.
- The coordinates of the point where the line crosses the y -axis are (0,1). The y -value of the point is 1.
- The coordinates of the point where the line crosses the x -axis are ($1\frac{1}{2}$, 0). The x -value of the point is $1\frac{1}{2}$.



The **intercept** is the value of a variable when all other variables in the equation equal 0. On a graph, the intercept is the value where a function crosses an axis.



Encourage students not to just put the lesser values on the left and greater values on the right, but instead to verbalize their answers. Scaffolding for this process can include sentence stems that reflect how students should read inequality sentences.



For auditory and kinesthetic learners, consider pairing actions with features. Examples:

- Every time someone says 'intercept' they must pretend to intercept a football pass*
- Every time someone says 'slope' the whole class must form a positive or negative slope with their arms*



Where else have you heard the word "intercept" before? Is the meaning similar? How many types of intercepts do you think a graph can have?



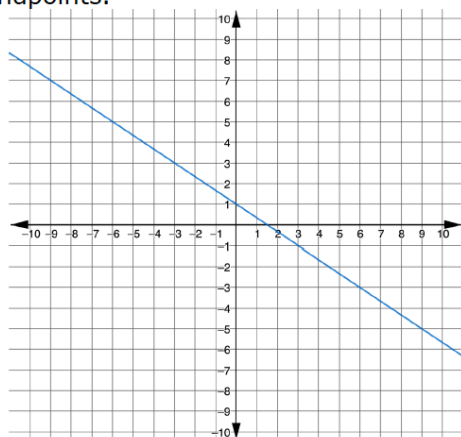
Intercepts can be expressed as an ordered pair or as a value. Students should be familiar with both.

4.2.2 Exploration: Key Features From a Graph (continued)

- h. Work with your partner to use the definition of **intercept** to determine the intercepts.

x-intercept	y-intercept
$1\frac{1}{2}$	1

2. A line that represents a function is shown on the coordinate plane. The line is similar to the line segment represented in the previous problem except it does not have endpoints.



Work with your partner to find the key features of the line.

Domain	Range	As x-values increase, y-values ...
All real numbers	All real numbers	Decrease
Slope	x-intercept	y-intercept
$-\frac{2}{3}$	1.5	1

3. Which key features were different for the line and line segment?

Domain / Range



Draw a line that has more than one x- or y-intercept. What type of shape could have more than one x- or y-intercept?



Consider which pieces of information in the table on domain and range can provide information for future remediation. Review student work in this table to determine areas that may require additional support. Some small group options could be:

- Identifying domain and range from a graph
- Identifying slope from a graph
- Identifying intercepts from a graph
- Interpreting meaning for key features



Why do you think both the domain and range are all real numbers? How would the line be different for the domain and range to be different? Explain.

4.2.3 Exploration: Horizontal and Vertical Lines

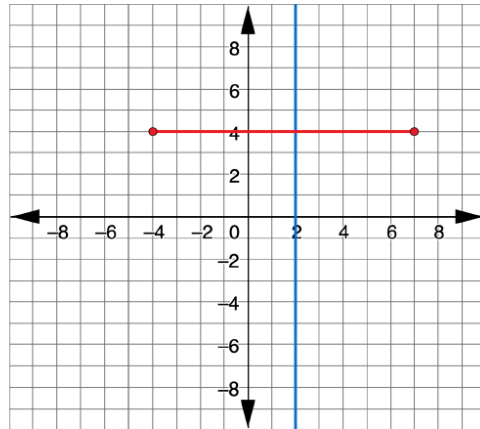
Component Overview: Students extend their exploration of key features of a graph with horizontal and vertical lines.



4.2.3 Exploration: Horizontal and Vertical Lines

- The graph shows a vertical line in blue and a horizontal line segment in red.

Work with your partner to find the key features for each.



Red Line Segment

Domain	Range	As x -values increase, y - values ...
$-4 \leq x \leq 7$	$y = 4$	Stay the same
Slope	x -intercept	y -intercept
0	none	4

Blue Line

Domain	Range	As x -values increase, y - values ...
$x = 2$	All real numbers	x -values do not increase
Slope	x -intercept	y -intercept
Undefined	2	None

- How does the range of a horizontal line relate to its y -intercept? *the range of a horizontal line has the same value as the y -intercept*
- How does the domain of a vertical line relate to its x -intercept? *the x intercept as the same value as the domain of a vertical line.*



This activity conceptually replicates the process from the previous activity, but positions students to grapple with different representations both on the graph as vertical and horizontal lines, as well as mathematically — now using an equals sign instead of an inequality symbol for domain or range. Students may not identify the importance and conceptual understanding of the difference between a slope of zero and a slope being undefined. Consider asking students one of the following questions so that students differentiate between slope being zero and slope being undefined:

- How is the slope of a horizontal line represented using $\frac{\Delta y}{\Delta x}$?
- How does $\frac{\Delta y}{\Delta x}$ lead to the slope of a vertical line being undefined?



For a challenge, have students determine three other equations that have a slope of 0 and three other equations that have an undefined slope. Ask them to categorize each equation under a header of what type of slope.

4.2.4 Wrap-Up: Reflection

Component Overview: Students will be completing a self-reflection based on what they learned in this lesson. Consider using quantitative data from Exploration activities to pair with this qualitative self-reflection.



4.2.4 Wrap-Up: Reflection

Complete each statement as you reflect on today's learning.

1. The concept that made the most sense to me today is...

Answers will vary.

2. One concept that I still do not understand from today is...

Answers will vary.



For students who struggle with writing, consider prompting the student to depict their meaning using models, drawings, or pictures.